

GAP IN SENSOR FUSION

However, the IAF faces a significant challenge in achieving seamless interoperability between its diverse fleet of fighter jets, including the Rafale and Su-30MKI. Airborne data links or sensor fusions are crucial for secure, real-time data exchange between aircraft, ground stations and airborne surveillance like satellites and AWAC (airborne early warning and control) aircraft to get a clear picture of the battlefield to form 'kill chains'—a crucial element in modern warfare. The lack of standardised data link protocols across these platforms hampers effective coordination and situational awareness, something faced during Operation Sindoor.

A specific issue arises with France's refusal to share the Rafale's source code. The code controls critical systems like the Thales RBE2 AESA radar and Modular Mission Computer (MMC). France's refusal, citing security, strategic and commercial concerns, limits India's ability to independently modify or upgrade the Rafale's systems. While Dassault has cooperated on integrating some Indian weapons such as the Astra Mk1 air-to-air missiles and SAAW (Smart Anti-Airfield Weapon), the lack of source code access forces India to rely solely on the manufacturer for further integration of the jet with new weapons or sensors.



"UNDER TRUMP, THE US IS UNPREDICTABLE. IT IS PUSHING 4TH-GEN JETS LIKE THE F-16 OR F-15E BEFORE OFFERING THE F-35. IF INDIA WANTS 5TH-GEN JETS AS A STOPGAP ARRANGEMENT, IT CAN GET THE SU-57"

AIR MARSHAL ANIL CHOPRA (RETD)
Former director-general, Centre for Air Power Studies

Air Marshal M. Matheswaran (retd), former deputy chief of Integrated Defence Staff, emphasises the need for rapid induction of new aircraft with technology transfer. "If India negotiates for additional Rafales, it should insist on access to the source code and technology-sharing. I don't think the F-35 should be negotiated, as it's very expensive and the Americans will never share any technology. We should try for the Su-57, as Moscow is looking for partners and will be happy to transfer technology," he adds.

Another key IAF official says that to complement its fighter jet fleet and enhance overall combat effectiveness, the IAF should focus on further developing capabilities in guided weapons such as surface-to-air missiles (SAMs), electronic warfare, space-based assets and unmanned aerial vehicles. Additionally, the deployment of long-range missiles like the BrahMos-2 would bolster the IAF's deterrence.

Christopher Clary, a non-resident fellow at the Washington DC-based Stimson Center, also feels that India can compensate for its slow pace of acquisition by emphasising other strengths, such as SAMs for air defence and cruise and ballistic missiles for strikes. "Even acknowledging those strengths, I think India will need more modern manned aircraft sooner rather than later," he says. Clary thinks buying more Rafales can give India economies of scale.

Air Marshal S.B.P. Sinha (retd), former deputy chief of the IAF, has a different take. He says buying new fifth-gen jets will require extensive negotiations, as they have unique maintenance and operational requirements. The current weapons with the IAF, he says, are unfit for use in the internal weapons bays of both the F-35 and the Su-57 in terms of length, width, weight or the ability to be released at supersonic speed. Therefore, a fifth-gen package needs to include weapons too. "We need to focus on VHF (very high frequency) radars to detect stealth and low RCS (radar cross section) aircraft, and develop requisite algorithms to control weapons against stealth aircraft using VHF radars," says Air Marshal Sinha. He adds that setting up plants for manufacturing 40-60 fifth-gen jets would exponentially increase the unit cost of each aircraft.

With its indigenous fifth gen fighter years away and regional rivals accelerating their own procurements, India has to make a bold, immediate decision. ■